

Streams



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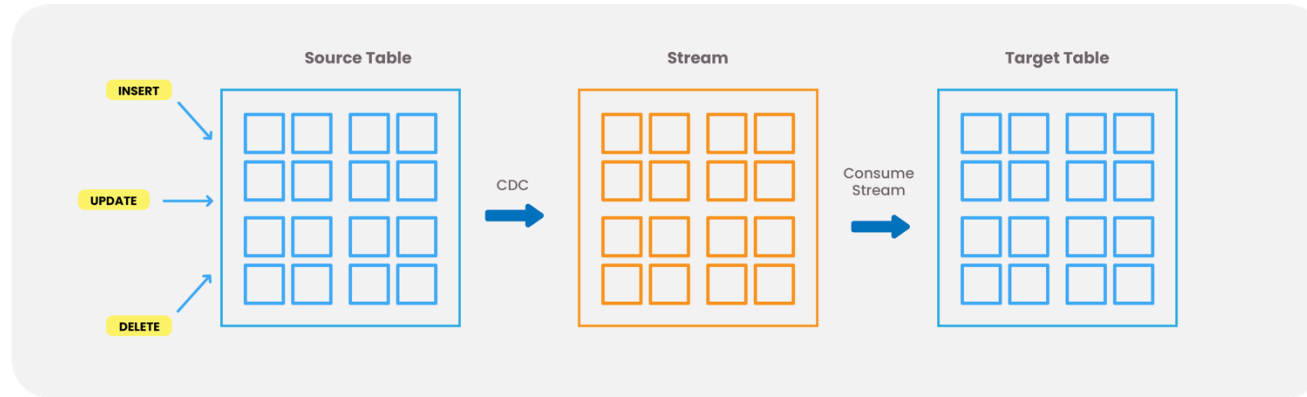
Cloud Data Architect
Snowflake Snowpro Certified

Section Introduction-Streams

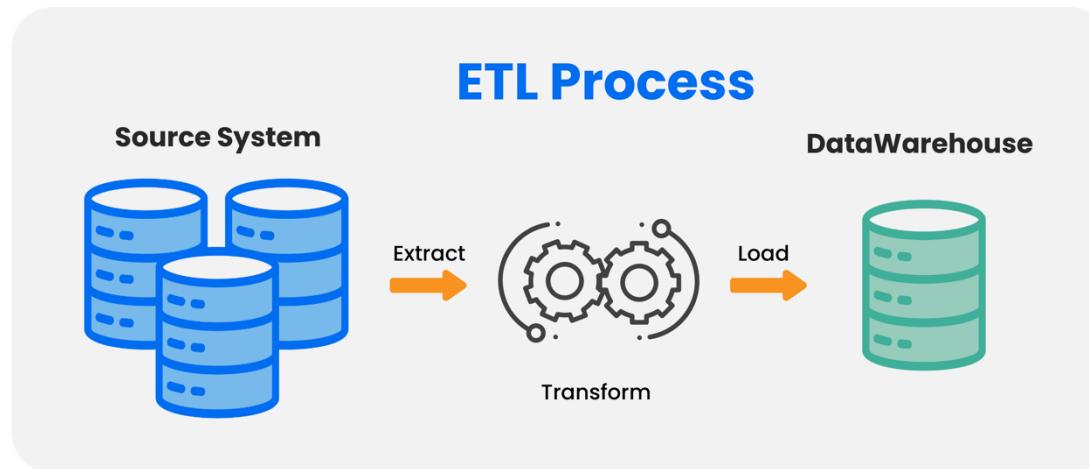
- What are Streams ? How they can be used to process data changes.
- Different type of Streams.
- LAB Streams
 - Create a STANDARD Stream.
 - Use Tasks with Stream to implement Change Data Capture(CDC).
 - Create a Stream on a view.
 - Create an APPEND-ONLY Stream.
 - How to empty a Stream.
 - Understand change tracking and how it is different from Streams.
- Summary.

What are Streams?

Streams are Snowflakes implementation of Near Real Time Change Data Capture(CDC) with the help of which we can capture data changes like INSERT,UPDATE and DELETE that are happening in a source table in a matter of few seconds and propagate these changes to target table.



Traditional ETL follows batch mode.



What are Streams?

Streams have 2 important metadata columns called METADATA\$ACTION and METADATA\$ISUPDATE.

METADATA\$ACTION = INSERT.

METADATA\$ACTION = DELETE.

METADATA\$ISUPDATE = TRUE.

METADATA\$ISUPDATE = FALSE



What are Streams?

Source table

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
100	STEVEN	KING	SKING	515.123.4567	2003-06-17	AD_PRES	24000.00	null	null	90
101	NEENA	KOCHHAR	NKOCHHAR	515.123.4568	2005-09-21	AD_VP	17000.00	null	100	90
102	LEX	DE HAAN	LDEHAAN	515.123.4569	2001-01-13	AD_VP	17000.00	null	100	90

Stream (INSERT/UPDATE)

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID	METADATA\$ACTION	METADATA\$ISUPDATE
100	STEVEN	KING	SKING	515.123.4567	2003-06-17	AD_PRES	21000.00	null	null	90	INSERT	TRUE
100	STEVEN	KING	SKING	515.123.4567	2003-06-17	AD_PRES	24000.00	null	null	90	DELETE	TRUE
102	LEX	DE HAAN	LDEHAAN	515.123.4569	2001-01-13	AD_VP	17000.00	null	100	90	INSERT	FALSE

Stream(DELETE)

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID	METADATA\$ACTION	METADATA\$ISUPDATE
100	STEVEN	KING	SKING	515.123.4567	2003-06-17	AD_PRES	21000.00	null	null	90	DELETE	FALSE

Different types of Streams?

There are 3 types Streams in Snowflake

- Standard (Default).
- Append-only (Inserts).
- Insert-only (External Tables).



- Standard Stream
Tracks all DML changes to the source object including INSERT, UPDATE and DELETE (including table truncates). STANDARD stream is also known as the DEFAULT Stream.
- Append-only
An Append-only Stream tracks table inserts exclusively. For Eg: If we insert 10 rows into a table with a stream and then update 5 of them, the Stream would capture data related to the inserted rows and would ignore the updates completely.
- Insert-only
Insert-only Stream applies to external tables alone. We can create Streams on external tables and rows from the new files added to the external table will be captured in the Stream.

Summary

- Streams are used to capture changes occurring in a source table like INSERT, UPDATE and DELETE and propagate them to the target table with the help of MERGE, CREATE TABLE or INSERT.
- 3 Types of Streams are supported by Snowflake
 - STANDARD
 - APPEND-ONLY
 - INSERT-ONLY.
- Standard Stream tracks all DML changes to the source object including INSERT, UPDATE and DELETE (including table truncates). STANDARD Stream is also known as the DEFAULT Stream.
- An APPEND-ONLY Stream tracks table inserts exclusively. APPEND-ONLY streams are more performant than STANDARD streams as they don't need to track updates and deletes.
- INSERT-ONLY Stream applies to external tables alone. New files added to the external table path will be captured by the Stream
- When a row is updated in the source table it creates 2 rows in the STANDARD Stream. 1 row with the old value of the row before the UPDATE and 1 row with the new value after the UPDATE to represent the before and after values.
- Streams have 2 important metadata columns `METADATA$ACTION` and `METADATA$ISUPDATE`
- `METADATA$ACTION` tells us what action was applied on the source data and can have the values of INSERT and DELETE.
- `METADATA$ISUPDATE` is a Boolean field which takes the values of TRUE or FALSE. When a row is inserted or deleted from the source table the value of `METADATA$ISUPDATE` is set to FALSE and if the row is updated in the source table `METADATA$ISUPDATE` is set to TRUE.

Summary

- Streams can be created on views and they help simplify Stream creation and management as they can be used to track changes happening on multiple tables without having to create a Stream on every table used in the view.
- Tasks and Streams are used with each other to schedule regular processing of Stream data.
- Tasks use `SYSTEM$STREAM_HAS_DATA('<Stream Name>')` to determine if there is data in the Stream and execute the Task SQL which is generally a MERGE.
- Data from a Stream once inserted into the target table is consumed and the Stream becomes empty.
- To empty a Stream re-create the Stream or consume data from the Stream by creating a table from it or Inserting data from the Stream to a target table. A Stream cannot be truncated to empty it.
- We can enable change tracking on a table by setting `CHANGE_TRACKING = TRUE`.
- Use the `CHANGES` clause with the `SELECT` to capture changes in the data within a time window.
- Data that is consumed from a table that uses `CHANGE_TRACKING = TRUE` and `CHANGES` clause does not get consumed and hence we can insert this data into multiple tables without emptying it.
- Data between a time interval can be captured from a table that has `CHANGE_TRACKING` enabled on it.

Script to end the section.

Hope you found this section on Streams exciting. If you have any questions please do reach out in the Q&A and I will respond back in a few days for sure if not sooner.

Before I let you go, I wanted to discuss a common question I get regularly asked with regards to Streams. In the real project world we would want to capture changes that are happening on a source system table which is outside of Snowflake and how will Streams help us with that if we are not able to create a Stream outside of Snowflake on the source system source table. This is a very good question.

We cannot create Streams outside Snowflake on Source system Tables. However we can use 3rd party tools like Oracle Golden Gate to capture INSERTS and UPDATES and DELETES and move them to a table in Snowflake. We can create a Stream on this table and now we are able to consume all the changes that are happening in the source system table to our target table in Snowflake. This is a question you might get in an interview so do remember this.

Hope you enjoyed this section as much as I enjoyed making it. Please join me in the next section and learn "How to snowflake".

Thank you



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